VidhyaSaathy

Learn • Grow • Together

Product Blueprint and Application Specification

Complete working document for the VidhyaSaathy app: every role, screen, feature, workflow and open decision.

Sky Blue	Peach	Warm White	Navy	Light Blue
#7FB3FF	#FFC8A3	#FFF9F5	#1E3A8A	#E6F0FF
Primary actions, active learning states	Supportive highlights, celebrations	Main background, comfortable reading	Headings, text, wordmark	Cards, panels, tinted areas

Working draft v1 | For team review | Vikrant University Junior Hackathon 2026

How to use this document

VidhyaSaathy is a school-connected learning workspace. Each school class gets an online workspace that mirrors the offline class, so the school's support continues after the bell rings. This document describes the whole product in one place, so a designer, developer, school stakeholder or new team member can understand it without further explanation.

Every feature has a stable ID such as **AUTH-03** or **AI-07**. Use the IDs in meetings and tickets. IDs and section numbers are clickable links throughout the PDF, and the sidebar bookmarks let you jump to any section.

STATUS	MEANING
CORE	Required for the central learning loop
NEXT	Valuable once the core loop works
ADVANCED	Needs extra engineering, content or validation
FUTURE	Longer-term possibility, kept for the record
UNDECIDED	Needs a team decision first

Items marked **OPEN** or **[OPEN]** are decisions the team has not made. Do not guess them. No feature is marked as built, because no working prototype was supplied. Checklists use **[ ]** for work still to do.

Contents

01	Document overview, sources and source conflicts
02	Product identity and philosophy
03	The core learning loop
04	Brand, logo, mascot and design system
05	User roles and permissions
06	Application map and screen flow
07	School setup and account creation
08	Sign-in, account recovery, shared devices and onboarding
09	Student experience
10	Classes, subjects, chapters and resources
11	Guided self-study sessions
12	Homework, assignments and assessments
13	AI learning engine (core)
14	Advanced and experimental AI features
15	Progress, learning history and motivation
16	Teacher experience
17	Teacher insights and analytics
18	Parent experience
19	School administration
20	Notifications and communication
21	India-specific layer
22	Complete cross-role user journeys
23	Feature dependency map
24	Conceptual data model
25	Security, privacy and responsible AI
26	Screen inventory and interface states
27	Error handling and edge cases
28	Implementation phases and status definitions
29	Acceptance criteria checklist
30	Open decisions and risks
31	Glossary
32	Final product definition

Feature index

Every feature ID in this document, with its status and the section where it is explained. Click an ID to jump to it.

ID	Feature	Status	Section
AUTH-01	School registration	CORE	07
SCH-01	School setup wizard (8 steps)	CORE	07
SCH-02	School profile	CORE	07
AUTH-02	Teacher account creation	CORE	07
AUTH-03	Student account creation	CORE	07
AUTH-04	Student PIN sign-in	CORE	07
AUTH-05	Parent account creation and linking	CORE	07
AUTH-06	Admin, teacher and parent sign-in	CORE	07
AUTH-07	Session management	CORE	07
AUTH-08	Manual approval fallback	NEXT	07
AUTH-09	Profile chooser on shared devices	NEXT	07
ONB-01	Student welcome and first-run onboarding	CORE	08
ONB-02	Teacher onboarding	CORE	08
ONB-03	Parent onboarding	CORE	08
ONB-04	Account recovery	CORE	08
ONB-05	Access-denied and not-yet-available states	CORE	08
STU-01	Student home screen	CORE	09
STU-02	Today's learning plan	CORE	09
STU-03	Continue card	CORE	09
STU-04	Subjects and chapters	CORE	09
STU-05	Notes and resources view	CORE	09
STU-06	Recordings and videos	CORE	09
STU-07	Assignments and quizzes list	CORE	09
STU-08	Ask Saathy	CORE	09
STU-09	My progress	CORE	09
STU-10	Profile, language and settings	CORE	09
STU-11	Class space	NEXT	09
STU-12	Help and feedback	CORE	09
STU-13	Learning games entry	NEXT	09
LES-01	Class workspace	CORE	10
LES-02	Resource upload	CORE	10
LES-03	Concept-tagged notes	CORE	10
LES-04	Publishing and notifications	CORE	10
LES-05	Note publishing from day one	CORE	10
LES-06	Chapter history	CORE	10
LES-07	Classroom recordings	CORE, if school allows	10

ID	Feature	Status	Section
LES-08	Curated videos (trimmed YouTube)	CORE	10
LES-09	Resource reuse	NEXT	10
LES-10	Curriculum mapping by board	NEXT	10
SES-01	Session structure	CORE	11
SES-02	Session player	CORE	11
SES-03	Session builder (teacher)	CORE	11
SES-04	Session completion rules	CORE	11
SES-05	Timed release	CORE	11
SES-06	Session reuse across years	NEXT	11
SES-07	AI-drafted session skeleton	ADVANCED	11
ASM-01	Student assignments list	CORE	12
ASM-02	Assignment and homework builder (teacher)	CORE	12
ASM-03	Answering and submission	CORE	12
ASM-04	Auto-marking	CORE	12
ASM-05	Teacher review queue	CORE	12
ASM-06	Assessments and tests	CORE	12
ASM-07	Quiz after each session	CORE	12
ASM-08	Question bank	CORE	12
ASM-09	Board-format practice	NEXT	12
ASM-10	Test readiness summary	NEXT	12
ASM-11	Assignment and session reminders	NEXT	12
AI-01	Context-grounded Ask Saathy	CORE	13
AI-02	Context retrieval	CORE	13
AI-03	Concept and prerequisite graph	CORE	13
AI-04	Mistake explanations	CORE	13
AI-05	Personalised explanation modes	CORE	13
AI-06	Personalised study plan	CORE	13
AI-07	Prerequisite gap detector	CORE	13
AI-08	Evidence and mastery estimate	CORE	13
AI-09	Adaptive practice	CORE	13
AI-10	Hints and scaffolding	CORE	13
AI-11	Socratic mode	CORE	13
AI-12	Doubt inbox	NEXT	13
AI-13	Concept map (student view)	CORE	13
AI-15	Honesty layer	CORE	13
AI-16	Ask Saathy safety settings	CORE	13
AI-17	Frustration-aware responses	NEXT	13
AI-18	Warm-up radar (forgetting detector)	ADVANCED	13
AI-19	AI feedback drafts for teachers	CORE	13

ID	Feature	Status	Section
AI-20	Natural-language check	NEXT	13
AI-21	Learning Remix (same concept, new context)	ADVANCED	14
AI-22	AI question and lesson drafts (teacher-approved)	CORE for drafts, ADVANCED for lessons	14
AI-23	Lesson compiler (one source, many outputs)	ADVANCED	14
AI-24	Concept radar for classes	ADVANCED	14
AI-25	Targeted reteaching assistant	ADVANCED	14
AI-26	What-If Machine (simulations)	ADVANCED	14
AI-27	Mistake Time Machine (solution tracing)	ADVANCED	14
AI-28	Homework autopsy	ADVANCED	14
AI-29	Mistake DNA (error categories)	NEXT	14
AI-30	Full-lesson search and AI segmentation	FUTURE	14
AI-31	Chapter-to-comic studio	FUTURE	14
AI-32	Branching learning adventure	FUTURE	14
AI-33	Concept court (debate between claims)	FUTURE	14
AI-34	Reverse exam mode (rubric feedback)	ADVANCED	14
AI-35	Viva simulator	ADVANCED	14
AI-36	Explain-it-back mode (Feynman style)	ADVANCED	14
AI-37	Reality scanner (everyday links)	FUTURE	14
AI-38	Past-and-future reasoning challenge	FUTURE	14
AI-39	Digital twin (predicted next test)	ADVANCED	14
AI-40	Teacher emphasis radar	ADVANCED	14
AI-41	Mistake autopsy view (class)	ADVANCED	14
AI-42	Teacher pattern profile ("what tends to be asked")	UNDECIDED	14
AI-43	Night recap (audio summary)	FUTURE	14
AI-44	Big-sibling tutor persona (name UNDECIDED)	UNDECIDED	14
PRG-01	Student progress view	CORE	15
PRG-02	Learning history	CORE	15
PRG-03	Achievement milestones	NEXT	15
PRG-04	Improvement badges (growth, not rank)	NEXT	15
PRG-05	Streaks with freeze	NEXT	15
PRG-06	Daily five-minute challenge	NEXT	15
PRG-07	Reflection prompts	NEXT	15
PRG-08	Parent-facing study prompts	NEXT	15
MOT-01	MCQ rush	NEXT	15
MOT-02	Match the concept	NEXT	15
MOT-03	Rapid revision	NEXT	15
MOT-04	Streak challenge	NEXT	15
MOT-05	Boss level (chapter unlock)	NEXT	15

ID	Feature	Status	Section
TCH-01	Teacher home	CORE	16
TCH-02	Class list and class detail	CORE	16
TCH-03	Roster	CORE	16
TCH-04	Publish notes and resources	CORE	16
TCH-05	Lesson editor	CORE	16
TCH-06	Session builder	CORE	16
TCH-07	Homework and quiz creator	CORE	16
TCH-08	Auto-marking and review queue	CORE	16
TCH-09	Class announcements	CORE	16
TCH-10	Teacher settings	CORE	16
TCH-11	Class teacher role	CORE	16
TCH-12	Reteaching decision	CORE	16
ANA-01	Class completion view	CORE	17
ANA-02	Concept heat map	CORE	17
ANA-03	Doubt clustering	ADVANCED	17
ANA-04	Intervention record and follow-up	ADVANCED	17
ANA-05	Mistake autopsy class view	ADVANCED	17
ANA-06	Class trend view	NEXT	17
ANA-07	Students who may need extra support	NEXT	17
ANA-08	Teacher mirror (teaching time versus interaction)	ADVANCED	17
PAR-01	Parent home	CORE	18
PAR-02	Child view	CORE	18
PAR-03	Progress summary (weekly or on event)	CORE	18
PAR-04	Assignments and deadlines	CORE	18
PAR-05	Support suggestions ("Ask this today")	NEXT	18
PAR-06	Linked children and account settings	CORE	18
PAR-07	Plain-text message fallback (WhatsApp or SMS)	NEXT	18
PAR-08	Privacy boundaries (hard rules)	CORE	18
PAR-09	Parent meeting preparation	ADVANCED	18
SCH-03	Academic year	CORE	19
SCH-04	Classes and sections	CORE	19
SCH-05	Subjects	CORE	19
SCH-06	Teacher assignments	CORE	19
SCH-07	Student enrolment	CORE	19
SCH-08	Parent linking management	CORE	19
SCH-09	Invitations	CORE	19
SCH-10	Admin home	CORE	19
SCH-11	Content library permissions	CORE	19
SCH-12	School settings	CORE	19

ID	Feature	Status	Section
SCH-13	Academic reports	NEXT	19
SCH-14	Admin activity log	CORE	19
COM-01	Student and class notifications	CORE	20
COM-02	Teacher announcements	CORE	20
COM-03	Parent notifications	CORE	20
COM-04	Reminders	NEXT	20
COM-05	Quiet hours	CORE	20
COM-06	Delivery failure	CORE	20
COM-07	Notification centre	CORE	20
COM-08	Direct messaging	NOT IN FIRST RELEASE	20
IND-01	Offline-first study and sync	NEXT, access requirement	21
IND-02	Data saver and audio-only mode	NEXT	21
IND-03	Language layers	CORE for UI, NEXT for AI	21
IND-04	Low-end device support	CORE	21
IND-05	Calendar awareness	NEXT	21
IND-06	Parent and student shared-device privacy	CORE	21

01 Document overview, sources and source conflicts

1.1 Purpose

VidhyaSaathy is a school-connected digital learning workspace. Each school class gets an online workspace that mirrors the offline class, so that the school's support continues after the school bell rings. This document turns the team's sources into one working specification that a designer, developer, school stakeholder or new team member can read without further explanation.

1.2 Sources reviewed

S1 Founder's voice-note transcripts (Hindi and Hinglish). Translated in this document.

S2 Founder's written master prompt on the product idea and research goals.

S3 "Team Guide" PDF, team draft v1 dated 13 October 2026. It marks each item as CONFIRMED, PROPOSED or OPEN.

S4 Master working spec with feature numbers VS-01 to VS-100 (the "Krazy features" list), including AI concepts, teacher tools, parent tools and India-specific layers.

S5 Product Vision and Feature document (long-form PRD style), with sections on student, teacher, parent and school workflows.

S6 Brand and theme image: logo, cloud mascot, palette, mascot expressions, a sample poster and a sample phone home screen.

S7 Earlier summary and feature-list PDF prepared for the team in English.

Not supplied: any working prototype, screenshots of a built app, school data, user research results, or technical test results. Nothing in this document is evidence that the product works.

1.3 Source conflicts the team must resolve

C1 Product name. Team Guide says "The name is not final" and the mockup is titled ClassOS. All other sources use VidhyaSaathy. Decision: 15 (Section 30).

C2 Tagline. The logo shows "Learn . Grow . Together". Earlier drafts proposed "Grow at your own pace." Only one tagline should appear publicly. Decision: 16.

C3 Student login. S3 says a roll number alone is not secure. S4 describes a roll-number search, a school-issued activation code and a parent-phone OTP. Direction in this document: the roll number identifies the student but never logs them in. Method is [OPEN].

C4 Technology stack. S3 lists options (React or Next.js, or Flutter;

Node.js or Django; MongoDB or PostgreSQL). S4 names Supabase with PostgreSQL, TypeScript and Tailwind. Decision: 14 (Section 30).

C5 Performance bands. S3 and S5 call them "four internal bands". S4 assigns them to a labelled "Student Progress" view in one place. This document follows S3: bands are never shown to students as labels.

C6 Session completion. S3 proposes that a session is complete only when a quiz or check is attempted. S4 suggests slides viewed or quiz passed.

Direction: a quiz or check attempt is required. Decision: 8.

C7 Teacher-controlled automation. S4 includes AI generation of questions, lessons and a predicted-question feature (AI-40). S2 and S5 require teacher approval. This document applies teacher approval everywhere AI content can reach students.

C8 Parent visibility of AI chats. S4 and S5 both say private AI chats are not visible to parents by default. No conflict; recorded for clarity.

1.4 Conventions used in this document

- "Student", "teacher", "parent" and "school admin" mean the roles in Section 05.
- "Class workspace" means the online space for one class (for example Class 9-A).
- "Session" means a guided self-study session unless stated otherwise.
- "Evidence" means a recorded learning action (for example, a quiz answer).
- "Mastery estimate" is a working estimate from evidence. It is never presented as a proven fact.

02 Product identity and philosophy

2.1 What VidhyaSaathy is

- A school-connected learning workspace for each class.
- A bridge between classroom teaching and home study.
- A place where teachers publish approved notes, recordings, sessions and homework.
- A place where students follow a clear next step after school.
- An AI helper that answers questions in the context of the school's own material.
- A progress view for teachers (class level) and parents (own child).

2.2 What VidhyaSaathy is not

- It is not a generic AI chatbot. The AI works inside the school's context.
- It is not a video library. YouTube videos are supporting material only.
- It is not a file-sharing app or a group chat.
- It is not a full school ERP. Transport, fees and attendance are out of scope unless the team approves them later.
- It does not replace teachers. Teachers approve content and make final assessment decisions.
- It does not claim to remove the need for coaching. The long-term goal is to make school support stronger; the effect is not yet proven.

2.3 Founder motivation

The founder was a strong student who scored above 90% in earlier classes.

In Class 9, studying became chaotic. The founder did not attend coaching, searched YouTube, asked friends for notes (and received conflicting versions), and used AI tools without a clear path. This is one student's experience. It is a reason to investigate the problem, not proof that every student faces it.

2.4 Product philosophy (ten principles)

P1 School remains central. The syllabus, teachers and approved resources form the base of the experience.

P2 AI supports teachers and does not replace them.

P3 Personalisation follows learning evidence, not labels. Students are never called weak, average or intelligent.

P4 Understanding matters more than activity. Opening slides is not learning.

P5 Parents receive useful insight, not only marks and not private chats.

P6 The product must work for real Indian school conditions: shared devices, several languages, variable connectivity and limited teacher time.

P7 The experience is calm and encouraging. No public shame-based rankings, no manipulative streaks, no excessive notifications.

P8 AI output must be grounded and reviewable. Important content should trace back to school material or another identified source.

P9 Privacy is a product requirement, not an afterthought.

P10 Every major feature connects to the learning loop: identify a need, take an action, check whether the action worked.

2.5 Long-term ambition

Over time the platform should connect the school's curriculum, its resources, its students' evidence and its teachers' interventions into one connected system, so that each class's support improves year on year.

03 The core learning loop

The whole product is built around one loop. Every feature in this document serves one or more steps.

1. Teacher teaches the lesson in class.
2. Teacher publishes notes, recording and a guided session to the class workspace (LES-01, LES-05, SES-01).
3. Student studies the session and attempts its checks (SES-02, SES-04).
4. The system records evidence: answers, time, hints, repeated errors (ASM-03, AI-08).
5. The AI identifies a possible gap and proposes one next step (AI-03, AI-06, AI-07).
6. Student receives the next step and tries a fresh question (AI-06, AI-09).
7. The mastery estimate updates (AI-08).
8. Teacher sees class-level patterns (ANA-01, ANA-02).
9. Teacher decides an intervention: reteach, assign practice or move on (ANA-04).
10. Parent receives an approved progress summary (PAR-03).
11. The loop repeats with the next lesson.

Worked example (Class 9 Physics, Motion, illustrative)

- 10:00 Teacher teaches Motion. Lesson recording is uploaded (LES-07).
- 14:00 Teacher publishes notes and creates Session 3, opening at 18:00.
- 18:30 Student A scores 5 of 5 and receives a harder Speed and Velocity set. Student B scores 2 of 5 on Acceleration and receives a simpler explanation with worked examples. Student C missed the lesson and receives a catch-up path.
- Next morning Teacher sees that most of the class is stuck on Acceleration and spends 10 minutes reteaching it.
- Evening Parent sees that Session 3 is complete, the quiz result and one suggested next step. The parent does not see Student B's private AI chat.

Note: names, scores and times are fictional and illustrate the workflow only.

04 Brand, logo, mascot and design system

Brand palette (from the team's theme image)

Sky Blue	Peach	Warm White	Navy	Light Blue
#7FB3FF	#FFC8A3	#FFF9F5	#1E3A8A	#E6F0FF
Primary actions, active learning states	Supportive highlights, celebrations	Main background, comfortable reading	Headings, text, wordmark	Cards, panels, tinted areas



Mascot expressions supplied by the team: Let's learn!, You got this!, Hmm..., Keep going!, Great job!, Time to rest.

Source: S6 (VidhyaSaathy_Logo_and_Theme.png). The image is the source of truth for the visual identity. Details below describe only what is visible or explicitly stated.

4.1 Logo

- Wordmark: "VidhyaSaathy" in a rounded, hand-lettered style, deep navy.
- Mascot: a friendly sky-blue cloud with a smiling face and peach cheeks, holding an open peach-and-white book. Small peach sparkle lines sit above the cloud.
- Tagline shown under the wordmark: "Learn . Grow . Together" (with small peach dots). This is the tagline in the logo.
- An app-icon version shows the cloud mascot on a white rounded square over a light blue background.
- A standalone "V" mark appears in the sample set: a navy V with a peach leaf or sprout above it. Its use is [OPEN] (see Decision 3 and Section 30).

Logo rules (proposed, for the design team to confirm):

[ ] Provide a light-background version, a dark-background version, a monochrome version and a small-size version.

[ ] Confirm whether the V mark is a secondary icon or an alternative logo.

[ ] Keep clear space around the wordmark equal to the height of the cloud.

4.2 Colour palette (hex values as shown in the supplied image)

```
Role              Name          Hex        Use
```

```
Primary           Sky Blue      #7FB3FF    Primary actions, active states,
```

key learning states

```
Accent            Peach         #FFC8A3    Supportive highlights, celebrations,
```

mascot book

```
Background        Warm White    #FFF9F5    Main background, comfortable reading
Text and strong   Navy          #1E3A8A    Headings, main text, logo wordmark
Surface           Light Blue    #E6F0FF    Cards, panels, tinted backgrounds
```

Usage ratio (from S6): about 80 percent clean interface, 15 percent soft colour, 5 percent mascot and illustration. Do not use the palette equally.

Accessibility check [ ]: confirm contrast of navy on warm white and navy on light blue meets WCAG AA for body text. Confirm sky blue is not used alone for text.

4.3 Typography

Supplied sources do not name fonts. The logo wordmark is a custom rounded style. The UI font is [OPEN]. Requirement: a clear, readable sans-serif with a comfortable body size on small phones.

4.4 Mascot expressions (six supplied states)

The mascot appears in six supplied expressions. Use them as defined:

*** Let's learn!** (waving, sparkle lines) Start of a session or greeting.

```
* You got this!    (happy, closed eyes)     Encouragement after progress.
* Hmm...           (question mark)          Thinking or confusion support.
```

*** Keep going!** (headphones, music note) Audio or recording activity.

```
* Great job!       (small heart)            Meaningful milestone.
* Time to rest     (sleeping, "zZ")         End of day or resting state.
```

Mascot behaviour rules (from S2 and S3):

1. Idle: slow, subtle floating motion only.
2. Greeting: short welcome when the student opens the learning space.
3. Thinking: restrained motion while the AI works.
4. Encouragement: small reaction after meaningful progress.
5. Confusion: appears when the student asks for help or repeatedly gets stuck.
6. Celebration: brief, only after a real milestone.
7. Resting: unobtrusive while the student reads or solves.
8. Error: says something failed, never blames the student.
9. Reduced motion: static image when the device requests reduced motion.
10. Consistency: the same recognisable character everywhere.

The mascot must never flash constantly, interrupt a lesson, or use emotional pressure to keep a student online. It does not control educational logic.

Voice, emotion detection and advanced character behaviour are not part of the first release.

4.5 Layout and interface style

- Student screens: clean, large touch targets, one primary action per screen.
- Teacher screens: denser information, same colours and typography.
- Parent screens: plain language, few numbers, one suggested action.
- Admin screens: structured tables and forms.
- Avoid purple or neon-heavy styling, cartoon-heavy layouts, heavy gradients and crowded cards. The sample home screen (S6) shows the intended level of decoration: three large cards, each with an icon, a title and one subtitle.

4.6 Sample home screen (from S6, for reference)

Greeting "Good morning, Learner!" with a small mascot, a search field ("What would you like to learn today?"), and three cards:

- My Classes: "View your subjects and chapters"
- Study Sessions: "Guided learning, made simple"
- Practice and Quizzes: "Test what you know" This is a visual reference, not a finished screen. The real home screen is specified in STU-01.

4.7 Empty, loading, success and error states (requirement)

- Empty: a short line, the mascot in a resting or "Let's learn!" pose, and one action. Example: "No session yet today. Your teacher will publish one soon."
- Loading: a skeleton of the card layout. Never a blank screen.
- Success: a short confirmation with a clear next step.
- Error: plain words, what happened, what the user can do next, and the mascot in "Hmm..." pose. Never blame the student.

05 User roles and permissions

5.1 Roles

Student	Studies, practises, asks the AI, views own progress.
Teacher	Manages assigned classes and subjects, publishes material, creates sessions and homework, reviews results.
Class teacher	A teacher with extra rights for one class: class updates, class space management.
Parent	Follows one or more verified children. Sees approved progress, not private AI chats.
School admin	Manages the school's structure: classes, sections, subjects, teacher assignments, enrolment and parent links.
Platform admin	Runs the platform itself: security, service health and authorised support. Separate from the school admin.

5.2 Permission principles

- Access is enforced on the server and in the database, never only hidden in the interface (see SEC-01).
- Changing a role label in the browser never grants access.
- A user sees only their school, their classes, their children or their own records.
- Teachers cannot see classes they are not assigned to.
- Parents cannot see children they are not linked to.
- Students cannot open teacher or admin pages by changing the URL.
- School admins manage structure. They do not see individual AI chats and do not rank students publicly.

5.3 Permission matrix (summary)

Action	Student	Teacher	Class T.	Parent	School A.
View own progress	yes	-	-	-	-
View own child's progress	-	-	-	yes	-
View assigned class progress	-	yes	yes	-	-
Publish notes and sessions	-	yes	yes	-	-
Submit homework	yes	-	-	-	-
Create school structure	-	-	-	-	yes
View private AI chats	-	-	-	-	-
Post class announcement	-	yes	yes	-	-

Note: "View private AI chats" is no for every role by default. Any future exception needs a written school policy and is [OPEN] (Decision 7).

5.4 Checklist

[ ] Role table reviewed by the team.

[ ] Permission matrix tested with the seeded demo accounts (Section 29).

06 Application map and screen flow

6.1 Top-level map

Public landing page

```
|-- Sign in ----------------------> Role check --> Student home
```

| \-> Teacher home

| \-> Parent home

| \-> School admin home

```
|-- Account recovery
|-- Accept invitation -----------> Role-specific onboarding
|-- Register a school -----------> School setup wizard (SCH-01)
```

6.2 Student map

Student home (STU-01)

```
|-- Today's plan (STU-02)
|-- My classes -> Subject -> Chapter -> Concept
|       |-- Notes and resources (LES-03)
|       |-- Recordings (LES-07)
|       |-- Videos (LES-08)
```

| \-- Sessions (SES-01)

```
|-- Assignments and quizzes (ASM-01)
|-- Ask Saathy (AI-01)
|-- Progress (PRG-01)
|-- Notifications (COM-01)
```

\-- Profile and settings (STU-10)

6.3 Teacher map

Teacher home (TCH-01)

```
|-- Class list -> Class detail -> Roster
|-- Resource library -> Upload (LES-02) -> Tag (LES-04) -> Publish
|-- Session builder (SES-03) -> Schedule -> Publish
|-- Assignment builder (ASM-02)
|-- Review queue (ASM-05)
|-- Class insights (ANA-01 to ANA-04)
|-- Announcements (COM-02)
```

\-- Settings (TCH-10)

6.4 Parent map

Parent home (PAR-01)

```
    |-- Child view (PAR-02)
    |-- Progress summary (PAR-03)
    |-- Assignments (PAR-04)
    |-- Support suggestions (PAR-05)
    |-- Notifications (COM-03)
```

\-- Account and linked children (PAR-06)

6.5 School admin map

Admin home (SCH-10)

```
    |-- School profile (SCH-02)
    |-- Academic year (SCH-03)
    |-- Classes and sections (SCH-04)
    |-- Subjects (SCH-05)
    |-- Teacher assignments (SCH-06)
    |-- Student enrolment (SCH-07)
    |-- Parent links (SCH-08)
    |-- Invitations (SCH-09)
```

\-- Content library permissions (SCH-11)

6.6 Transition rules

- After sign-in, the user lands on the home screen for their role only.
- Every list item opens a detail screen with a single back path.
- Every screen that contains learning content has one primary action.
- Session player (SES-02) is a focused full-screen mode. It hides the main navigation until the session ends or is saved and left.
- Leaving a session mid-way saves progress and returns to the home screen with a "Continue" card (STU-03).

END OF PART A (sections 01 to 06)

07 School setup and account creation

Schools need controlled onboarding because the platform holds student records.

AUTH-01 School registration — CORE

What: an authorised school representative registers the school.

How it works:

1. Representative starts registration from the landing page.
2. Enters school name, board (for example CBSE, ICSE or state board), city and administrative contact details.
3. Platform verifies the representative. Method [OPEN].
4. School profile is created. The representative becomes the first school admin.

Visible output: a confirmation and the school setup wizard (SCH-01).

Failure: duplicate school name in the same city shows a warning and asks the representative to confirm. Unverified representative sees a pending screen; no school data is shown.

For the prototype: a preconfigured demo school is used instead (Section 29).

SCH-01 School setup wizard (8 steps) — CORE

The wizard runs once, then can be revisited from settings.

Step 1 School profile (name, board, city, logo)

Step 2 Academic year (start and end dates)

Step 3 Classes and sections (for example 9-A, 9-B)

Step 4 Subjects per class (for example Physics, Maths)

Step 5 Teacher assignments (teacher x subject x class)

Step 6 Student list (CSV upload or manual entry; see SCH-07)

Step 7 Parent contact details attached to each student Step 8 Activation codes generated; school downloads a distribution sheet; each class workspace goes live Each step saves on its own, so the wizard can be left and resumed.

Validation: missing required fields, duplicate roll numbers and invalid phone numbers are listed before the step can be completed.

SCH-02 School profile — CORE

Name, board, address for records, logo and contact. Edited by the school admin only.

AUTH-02 Teacher account creation — CORE

1. School admin assigns the teacher to subjects and classes (SCH-06).
2. Teacher receives an invitation (SCH-09).
3. Teacher opens the invitation, sets a password and confirms identity.
4. Teacher completes a short profile.
5. Teacher sees only the classes and subjects assigned to them.

Rule: a teacher never sees unassigned classes or student records.

Failure: expired invitation shows "Ask your school admin to resend."

AUTH-03 Student account creation — CORE

Students should not need a consumer-style sign-up if the school has enrolled them.

1. School enrols the student (SCH-07) and generates an activation code.
2. Student (or a parent helping) enters the school, roll or admission number, class and section.
3. System finds the matching enrolment record but does not reveal personal details.
4. Student enters the one-time activation code.
5. A one-time verification code goes to the parent's registered phone number (see AUTH-05). Method is [OPEN] (Decision 2).
6. Student sets a 4-digit PIN for daily sign-in (AUTH-04).
7. Student sees the welcome screen (ONB-01).

Rules:

- An arbitrary person cannot join a class by knowing the class name.
- Codes are one-time use and expire after a school-set period [OPEN].
- Five wrong code attempts lock the activation for 15 minutes.

Fallback: if the parent phone is unreachable, the school admin can approve the student manually (AUTH-08).

AUTH-04 Student PIN sign-in — CORE

After activation, the student signs in with their PIN. The roll number is never a password.

Rules: three wrong PINs trigger a short wait and a prompt to ask a parent or teacher to reset it. PIN reset requires parent or teacher approval.

AUTH-05 Parent account creation and linking — CORE

1. The student's enrolment already contains the parent's phone number.
2. When the student activates, the parent receives a message: "Your child has joined VidhyaSaathy. Download the app and sign in with this number."
3. Parent verifies the phone number with a one-time code.
4. Parent confirms the child's name, class and section.
5. The link is active. The parent sees only this child's approved data.

Multiple children: one parent account can hold several verified links and switch between them (PAR-06).

Guardians (uncle, aunt, grandparent) are common in India. The school admin can register a guardian as the parent contact, subject to school policy [OPEN].

Rule: a parent cannot connect to a child by searching a name or a list.

AUTH-06 Admin, teacher and parent sign-in — CORE

Email or school-issued ID with password for staff and parents. Two-step verification is optional for admins [OPEN]. Password reset sends a time- limited link to the registered email or phone.

AUTH-07 Session management — CORE

- Session expiry returns the user to the sign-in screen with a short message.
- Sign out ends the session on that device.
- Shared-device reminder: after sign-out on a shared device, the app shows the profile chooser (AUTH-09).
- Protected pages reject unauthorised requests on the server.

AUTH-08 Manual approval fallback — NEXT

School admin reviews a pending student or parent and approves or rejects with a reason. Used when OTP delivery fails or the school policy requires it.

AUTH-09 Profile chooser on shared devices — NEXT

Several family members can share one phone. The chooser lists the profiles on that device. Each profile has its own PIN. Switching takes two taps.

An interrupted session resumes where the user left off.

08 Sign-in, account recovery, shared devices and onboard

ONB-01 Student welcome and first-run onboarding — CORE

Steps:

1. Welcome screen with the cloud mascot ("Let's learn!").
2. Confirm class and section.
3. Choose a preferred explanation language if offered (IND-03).
4. Show today's first assigned activity.
5. Explain how to ask Saathy a question and how to ask for a different explanation.
6. Start learning.

The whole flow should take under two minutes.

ONB-02 Teacher onboarding — CORE

Show assigned classes and subjects, a short guide to publishing a resource, a guide to creating a session, and a demo class and demo lesson in demo mode.

ONB-03 Parent onboarding — CORE

Confirm the linked child, explain progress and notifications, explain the difference between progress and private student chats, and show one useful first action.

ONB-04 Account recovery — CORE

- Staff and parents: reset link by email or phone.
- Students: PIN reset by a parent or teacher (AUTH-04).
- Recovery never reveals whether an account exists to a stranger.

ONB-05 Access-denied and not-yet-available states — CORE

Shown when a user opens a page they cannot access. The message says the page is not available for the account, with a button to go home. No student data is shown.

Landing page (LAND-01) [CORE]

Sections in order: hero with the school-to-home promise; use cases for student, teacher, parent and school; a short explanation of the learning loop; a preview of sessions and AI help; a note on how school material grounds AI answers; privacy and child-safety information; a school contact or onboarding button; sign-in link.

The landing page must not claim proven learning results, existing school partnerships or capabilities that are not built.

09 Student experience

The student's question on every screen is: "What do I study now, and what do I do after?"

STU-01 Student home screen — CORE

Top to bottom:

1. Greeting with the mascot ("Good morning, Learner!" or the student's name).
2. Next-action card: one clear task, for example "Session 4, Motion, 15 minutes. Start?" Only one primary action appears here.
3. Today's plan (STU-02).
4. Continue card: the last activity, one tap to resume (STU-03).
5. My concepts map (AI-13 preview), colour-coded.
6. Teacher announcements.
7. Upcoming assignments and tests.

Ordering rule: an item due today or a missed-class recovery moves to the top. The screen never shows more than one primary action.

Empty state: "Nothing due today. Explore your subjects." with a subject link.

STU-02 Today's learning plan — CORE

A short list built from teacher-assigned tasks, deadlines and recovery suggestions.

- Teacher-assigned items are marked as required.
- AI-suggested items are marked as optional.
- The AI never changes a teacher's deadline and never removes an assigned lesson.

Example: "Required: Session 3 (Motion). Optional: 2 Speed practice questions."

STU-03 Continue card — CORE

Shows the last unfinished item: the session, slide or question the student left. One tap resumes at the same place. Saved progress is stored every few seconds and when the student leaves.

STU-04 Subjects and chapters — CORE

Subject list for the class. Each subject opens to its current chapter, all chapters in order, and a status per chapter: not started, in progress, completed. Past chapters remain open all year (LES-06).

STU-05 Notes and resources view — CORE

Shows notes, PDFs, presentations, worksheets and diagrams for a chapter or concept, each labelled with the subject, chapter and source. See LES-03.

STU-06 Recordings and videos — CORE

Shows recordings and approved videos attached to a concept. See LES-07 and LES-08.

STU-07 Assignments and quizzes list — CORE

Lists assignments with due date, status (not started, submitted, reviewed) and the teacher's name. See ASM-01.

STU-08 Ask Saathy — CORE

The AI question screen. See AI-01 to AI-05.

STU-09 My progress — CORE

Progress by subject and concept, recent improvements and one suggested next step. See PRG-01.

STU-10 Profile, language and settings — CORE

Name and photo (school-controlled where needed), preferred language (IND-03), reduced motion setting (UI-04), sign-out and PIN change.

STU-11 Class space — NEXT

A shared page where students and teachers of the same class see each other (names and profile photos only). The class teacher posts updates here.

No free chat between students in the first release [OPEN] (see COM-08).

STU-12 Help and feedback — CORE

Any screen has a help link. "Ask my teacher" sends a named doubt to the teacher (AI-12). A feedback button sends a short note to the platform team.

STU-13 Learning games entry — NEXT

A games card on the home screen lists the games unlocked for the current chapter (MOT-01 to MOT-05, Section 15).

10 Classes, subjects, chapters and resources

The hierarchy used everywhere in the product:

School > Academic year > Class > Section > Subject > Chapter > Concept > item (note, session, question, recording, video).

LES-01 Class workspace — CORE

Each class has its own workspace, for example Class 9-A. It contains the subjects taught to that class, the teachers for each subject, the class space (STU-11), and every published item in order of chapter.

Example path: Class 9-A > Physics > Motion > Distance and Displacement > Notes, Session 3, Recording segment, Homework, Quiz results.

Rule: the workspace is an organised environment, not a group chat or a file folder.

LES-02 Resource upload — CORE

Teacher uploads PDF, slides, worksheet, text or an approved link. Steps: 1. Choose class, subject and chapter. 2. Choose file or paste link. 3. System reads the file and shows a preview (text extraction status). 4. Teacher confirms the title and visibility. 5. Publish, or save as draft. Failure: unreadable file is kept as an upload, flagged "Preview not available," and can still be published (students open the original file). Limits (size and types) are [OPEN] and depend on the hosting plan.

LES-03 Concept-tagged notes — CORE

The key feature of the notes system is tagging, not uploading. Each note is linked to one or more concepts. When a student opens "Acceleration," the teacher's note for that concept appears in two taps. Students do not need to find the right note among friends' versions.

Rule: a note without a concept tag is still visible under its chapter but is not linked to the concept map.

LES-04 Publishing and notifications — CORE

Publishing makes the item visible to the authorised class. Students and parents may receive a notification (COM-01). Unpublishing hides the item but keeps student history intact.

LES-05 Note publishing from day one — CORE

Teachers can publish notes topic by topic as a chapter is taught, so the class workspace has material from the first day of school. This is the basis of "learning continues after the bell."

LES-06 Chapter history — CORE

Nothing is deleted from the chapter history during the academic year.

Students can reopen any earlier chapter, session and quiz. Teachers can archive items but cannot erase what students already completed.

LES-07 Classroom recordings — CORE, if school allows

Teacher uploads a recording of the lesson. Requirements: * The school enables recordings in school settings (SCH-12). Default is off until the school decides. * Recording is shown only to the class it belongs to. * Teacher can mark concept segments (start and end time) on the recording. * Students can watch the whole recording or only a segment. * Audio-only playback is supported (IND-02). Segmentation can be manual or AI-assisted. AI-assisted segment suggestions must be checked by the teacher before publishing (AI-30). Consent and privacy rules for recordings that include students are [OPEN] (Decision 13).

LES-08 Curated videos (trimmed YouTube) — CORE

Teacher pastes a video link, sets start and end times, tags it to a concept and adds a one-line reason. Students see the segment, for example "For this concept: 8:20 to 15:40." YouTube is supporting material, not a replacement. Rule: opening or playing a video does not count as understanding. A check is needed (AI-08).

LES-09 Resource reuse — NEXT

A resource can be reused in a later year or for another class. Reuse copies the link to the original, so updates made by the teacher appear in both places [OPEN: copy or link behaviour].

LES-10 Curriculum mapping by board — NEXT

At setup, the school selects its board. The platform offers a starting chapter and concept structure for that board, based on public curriculum listings. Teachers can edit it. The mapping is per board, so the same chapter may have different depth for different boards.

11 Guided self-study sessions

A guided session is the central student activity. It is a school-prepared learning experience, not a long PDF and not an unrestricted list of links.

SES-01 Session structure — CORE

Each session follows a fixed structure. The teacher can change the questions per class but not the structure.

1. Session introduction: what we will learn today.
2. Learning objective (one sentence).
3. Explanation slides: one small idea per slide.
4. Diagram or visual.
5. Solved example, step by step.
6. Optional recording segment or curated video.
7. Practice questions (three or four).
8. Mini quiz (about five questions).
9. Recap.
10. Next step, for example "Speed and Velocity."

Example: Session 3, Physics, Motion (fictional).

Slide 1: definition of distance (one idea). Slide 2: diagram of a path and a straight line. Slide 3: worked example. Then a 6-minute recording segment, five MCQs, a recap and the next step.

SES-02 Session player — CORE

Student — flow: 1. Student opens the session when it is available. 2. Intro screen shows objective and time estimate. 3. Student moves through slides with a Next button. A progress bar shows position. 4. Questions appear inside the flow. The student must attempt a question before moving past it. 5. Hints are available (AI-11). Hints do not reveal the full answer on the first request. 6. Quiz at the end, scored and recorded. 7. Summary screen: what is complete, the score and the next step. Rules: * Slides unlock one after another (SES-04). * Leaving mid-session saves progress (STU-03). * Session is complete only after its quiz or check attempt (SES-04).

SES-03 Session builder (teacher) — CORE

Teacher — steps: 1. Create a session from the template in SES-01. 2. Add slides, examples and questions. Drag to reorder. 3. Attach notes, a recording segment or a curated video. 4. Set availability: for example "Opens today at 18:00." Set a deadline if needed. 5. Adjust questions for a specific class or section, keeping the structure. 6. Preview the student view. 7. Publish. Reuse (SES-06): next year, the teacher copies the session, updates dates and questions, and publishes.

SES-04 Session completion rules — CORE

- Slides unlock sequentially. A slide's question must be attempted before the next slide opens (teacher can set a lighter rule for a class).
- A session is complete only when its quiz or check attempt is recorded.

Scrolling through slides does not complete a session.

- Completion creates an evidence record (AI-08).

Open: whether the teacher can allow completion by slides alone (Decision 8).

SES-05 Timed release — CORE

Teacher picks a release date and time. The system stores it in the school's time zone. Before release the student sees "Opens at 18:00 today." At release the session appears on the home screen. Deadlines are separate from release time and follow school policy. Release timing does not block learning arbitrarily. It only controls when the teacher's assigned work becomes visible.

SES-06 Session reuse across years — NEXT

Copy, update dates and questions, publish. Reuse should take about five minutes for an existing session.

SES-07 AI-drafted session skeleton — ADVANCED

Teacher uploads an existing presentation. The system drafts a session outline: slide breaks, definitions, practice question drafts. The teacher edits everything and publishes. No AI draft reaches students without teacher review (AI-22).

12 Homework, assignments and assessments

ASM-01 Student assignments list — CORE

Each assignment shows its title, subject, due date, status (not started, in progress, submitted, reviewed) and feedback status. Overdue items are shown in calm language, not red alarms.

ASM-02 Assignment and homework builder (teacher) — CORE

Teacher creates an assignment with: * Instructions and optional resource links. * Questions: multiple choice, multiple select, short answer, numerical, matching, or a question from the question bank (ASM-08). * Due date and allowed attempts. * Feedback release rule: immediately, after the due date, or after teacher review. * Assigned classes (only authorised classes appear). Teachers can also assign homework that stays on paper. Handwritten work remains valid. The app records only what is submitted in it.

ASM-03 Answering and submission — CORE

Student answers each question, reviews, and submits. The system confirms submission with a time stamp. If the connection fails, the answers stay on the device and submit when the connection returns (IND-01). A failed submission is never silently lost (Section 27).

ASM-04 Auto-marking — CORE

Objective questions (MCQ, matching, numerical with a fixed answer) are marked automatically. The student sees their score and, where the teacher allows, the correct answer and an explanation.

ASM-05 Teacher review queue — CORE

Short answers and written work appear in a review queue. Each item shows the student's answer, the marking guide and an optional draft feedback suggestion from AI (AI-19). The teacher edits, approves and publishes feedback. The queue shows borderline items first.

Rule: AI feedback is a draft only. The teacher's grade is authoritative.

ASM-06 Assessments and tests — CORE

Teacher-created tests in the same system as homework, with timed mode where justified. Teacher controls when answers and feedback become visible.

For high-stakes assessments, AI does not decide final grades. Human review is required [OPEN: which assessments count as high-stakes].

ASM-07 Quiz after each session — CORE

Each session ends with a quiz (SES-01 item 8). The quiz results go to the student's concept map (AI-13) and the teacher's class insights (ANA-01).

ASM-08 Question bank — CORE

A school-owned library of approved questions, tagged by class, subject, chapter, concept, difficulty (L1 basic, L2 medium, L3 application) and format (MCQ, assertion-reason, case-based, numerical, short answer). Built from teacher-written questions and approved AI drafts. Grows each year.

ASM-09 Board-format practice — NEXT

A practice mode that shows questions in the same format as the board exam.

Knowing a concept and practising the exam format are separate skills.

ASM-10 Test readiness summary — NEXT

Before a chapter test the student sees a plain summary, for example: "Strong: 4 of 6 concepts. Needs attention: Acceleration. Suggested: 30 minutes of focused revision tonight."

Rule: the summary never predicts a score or a grade. It names concepts only.

ASM-11 Assignment and session reminders — NEXT

Reminders for due or released work (COM-04).

13 AI learning engine (core)

The AI is the personalisation layer. Its core functions are: answer questions in the school's context, explain in more than one way, find the likely gap behind a wrong answer, and suggest one next step. Every AI feature follows the grounding and review rules in Section 25.

Testing rule (from S4): "Could a general chatbot do this on its own?" If yes, the feature is not a differentiator. The differentiators are the school's notes, the class's quiz results, the curriculum position and the student's learning history. Those exist only inside the platform.

AI-01 Context-grounded Ask Saathy — CORE

Input the AI receives: the student's class, board, subject, chapter, current lesson, relevant teacher notes, and recent quiz evidence where it helps.

Example: the student types "Explain acceleration." The AI answers in the teacher's notation and ends with "This comes from your teacher's notes, Motion, slide 4."

Rules:

- If the school material does not cover the question, the AI says it is giving a general explanation.
- The AI never says "your teacher said" unless the material shows it.
- Every answer ends with a short check for understanding.

Failure: if the AI service is unavailable, the screen shows the saved notes for that concept and a retry button (Section 27).

AI-02 Context retrieval — CORE

Before answering, the system selects the approved material for the student's class and concept. Only that material plus the question is sent to the AI service. The selection uses the concept tags from LES-03. Full search across all uploads is an advanced upgrade (AI-30).

AI-03 Concept and prerequisite graph — CORE

The platform stores which concepts depend on which. Example chain:

Distance > Speed and Velocity > Acceleration > Graphs of motion.

Teachers can review and correct these links.

AI-04 Mistake explanations — CORE

When a student answers wrongly, the system explains the likely reason in two or three sentences and shows the relevant slide or note. The student can ask for another explanation (AI-05).

AI-05 Personalised explanation modes — CORE

The student can ask for the same concept in another form:

- Simple language.
- Step-by-step.
- Everyday analogy.
- Worked example.

- Formal textbook wording (the form used in exams).
- Bilingual or Hinglish explanation (IND-03).

After each explanation the system checks whether understanding improved.

If the student is still stuck after two attempts, the system suggests teacher help rather than repeating the same answer.

AI-06 Personalised study plan — CORE

Trigger: the student says what is hard, for example "I am not able to understand Coordinate Geometry."

Steps the plan can include:

1. Revise the prerequisite if the evidence shows a gap (AI-07).
2. Read the teacher's note for the concept.
3. Watch a recording segment or curated video.
4. Study one worked example.
5. Solve three easy questions.
6. Take a five-question mini check.
7. Move to application questions if the check passes; repeat step 2 or 3 if it does not.

Each step links to a real school resource. The plan changes after each result. Teacher-assigned work stays visible and takes priority.

AI-07 Prerequisite gap detector — CORE

Logic, in plain words: when a student gets a later concept wrong, the system checks the earlier concepts on the chain.

- If the earlier concept is solid, the gap is in the later concept.

Recommend revision of the later concept.

- If the earlier concept is weak, move one step back and check again.
- Stop at the first weak link and recommend fixing it, for example: "Acceleration needs velocity first. Spend 10 minutes on velocity, then acceleration will feel easier."

Guardrail: if there is not enough evidence (for example, velocity was never practised), the system says so and recommends practice or a teacher question. It does not guess.

Minimum evidence before a prerequisite gap is suggested: [OPEN] (Decision 4).

AI-08 Evidence and mastery estimate — CORE

The system records:

- Each attempt: question, answer, correctness, time, hints used.
- Each session completion (SES-04).
- Teacher-confirmed status where a teacher has reviewed the concept.

From these, it keeps a working estimate per concept. Rules:

- Estimates are per concept, not one overall score (see Decision 5).
- One correct answer does not declare mastery. The estimate needs varied evidence, for example different question formats, spread over time.
- The system shows uncertainty. A low-evidence concept is shown as "Not enough practice yet," not as a low score.

- Activity, correct answers, concept demonstrated, teacher-verified and AI-inferred difficulty are kept separate (see Section 25.6).

AI-09 Adaptive practice — CORE

Practice difficulty moves between L1, L2 and L3 questions:

- Three correct answers in a row move the student up one level.
- Two wrong answers in a row move the student down one level and show an explanation.

The student stays in the zone where effort is visible. Rules are simple and visible to the teacher. No hidden difficulty tricks.

AI-10 Hints and scaffolding — CORE

Hint sequence for a practice question:

1. A pointer to the concept.
2. A guiding question.
3. The first step.
4. A worked example of a similar question.
5. The full explanation, when appropriate.

Teachers set hint rules separately for practice and for tests. In a formal test, the default is no hints.

AI-11 Socratic mode — CORE

Purpose: the AI guides the student to the answer instead of giving it, to protect learning and reduce copying. Example:

Student: "Solve this numerical."

AI: "What is given, and what do you need to find?"

Student: "Distance and time are given. I need velocity."

AI: "Which formula links velocity, distance and time?" A "tell me directly" option exists. Teachers can set Socratic mode as the default for homework-type questions.

AI-12 Doubt inbox — NEXT

If the AI answer does not satisfy the student, a "Ask my teacher" button sends the question to the teacher with the student's name. The teacher replies in the app. The AI filters basic doubts; real problems reach the teacher.

AI-13 Concept map (student view) — CORE

A tree of the chapter's concepts. Each node's colour shows the student's current status, using friendly labels:

- Mastered (green-tinted node)
- Almost there (sky-blue node)
- Let's fix this (peach node)
- Not started (grey node)

Locked concepts (prerequisites not yet solid) show a lock and a short reason. The student can see why something is locked. This supports planning by the student.

AI-14 Four internal performance bands (PROPOSED)

Internal routing ranges, never shown as labels:

0 to 25 percent	large gaps: simpler explanation, prerequisite revision
25 to 50 percent	basics partly clear: worked examples, guided practice
50 to 75 percent	basics clear: application questions, more practice 75 to 100 percent consistent understanding: next concept, challenges Rules: * Bands are a routing device, not a measure of intelligence. * Thresholds are not validated and must be tested with real data. * Tracking is per concept (AI-08). Bands apply to each concept's estimate.

AI-15 Honesty layer — CORE

Rules applied to every AI feature:

- Cite the source: "your teacher's notes" or "general explanation."
- Say "I do not know" or "the material does not cover this" when true.
- Do not invent citations, page numbers or teacher statements.
- Mark uncertain or sensitive answers for teacher review.
- Never give medical, psychological or permanent-ability conclusions.

AI-16 Ask Saathy safety settings — CORE

Age-appropriate language, topic limits to the school subject list, and polite redirection for off-topic requests. Details in Section 25.

AI-17 Frustration-aware responses — NEXT

The system notices interaction patterns only: the same question asked three times, five wrong answers in a row, or repeated hint requests. When this happens, the reply changes tone and pace, for example: "No problem.

This one is tricky. Let's start from the basics. What do you already understand?"

Rule: no camera, microphone or biometric input is used. The feature uses interaction data only and is switched off in the student's privacy settings if the school disables it [OPEN].

AI-18 Warm-up radar (forgetting detector) — ADVANCED

Before a session the student gets a three-minute warm-up of questions on concepts they are likely to have forgotten. The likelihood comes from the time since the last attempt and the score at that time (a simple spacing rule). Warm-up items come from the question bank and the student's own history. The student does not need a separate revision session.

AI-19 AI feedback drafts for teachers — CORE

For written answers the AI drafts feedback, for example "Correct direction, unit missing." The teacher edits and sends. Feedback drafts are never sent to students without teacher approval.

AI-20	Natural-language check	NEXT

A question typed in Roman Hindi, Hinglish or English is understood and answered in the language the student chooses (IND-03).

14 Advanced and experimental AI features

Most features in this section are ADVANCED, FUTURE or UNDECIDED (AI-22 is CORE for teacher-approved drafts only). None of them is part of the first release unless the team says so (Section 28). Each one needs a validation plan before any claim is made about it.

AI-21 Learning Remix (same concept, new context) — ADVANCED

The same concept is rewritten around the student's interests, for example cricket, bikes or cooking. Difficulty, formula and marking stay the same.

Example:

Standard: "A car goes from 0 to 20 m/s in 5 s. Find acceleration."

Remix: "A bowler's delivery goes from 0 to 140 km/h in 0.4 s. Find acceleration. Hint: convert to m/s first."

Rules: the teacher approves a sample of remixes (for example one in ten) before they are used. Students can report a remix that feels wrong or off-putting. Interest data is collected only if the student chooses it.

AI-22 AI question and lesson drafts (teacher-approved) — CORE for drafts, ADVANCED for lessons

Teacher selects notes and asks for ten MCQs. The AI drafts them. The teacher approves, edits or rejects each one. Approved items enter the question bank (ASM-08). AI content never goes directly to a student.

AI-23 Lesson compiler (one source, many outputs) — ADVANCED

From an uploaded lesson the AI proposes a structured lesson, simplified and detailed explanations, a concept map, prerequisite concepts, practice questions, misconception checks, a revision sheet, a session draft and a parent-friendly summary. Each output is a draft. Teacher review is required.

Outputs trace to their source where possible.

AI-24 Concept radar for classes — ADVANCED

Shared misunderstandings detected from assessment answers. Example: many students confuse distance and displacement, or choose the wrong formula, or calculate correctly but cannot explain the result. The teacher sees a concept-level view with frequencies (ANA-02). Where one wrong answer could have several causes, the view shows uncertainty.

AI-25 Targeted reteaching assistant — ADVANCED

When the radar shows a shared gap, the AI drafts a short explanation, a worked example, a diagnostic question and a five-minute activity. The teacher approves and decides whether to use it. A follow-up check tests whether the gap improved.

AI-26 What-If Machine (simulations) — ADVANCED

Students change one condition and predict the result before it is shown.

Example: "If distance increases but time stays the same, what happens to speed?"

Rules:

- Numbers come from deterministic formulas, never from free AI text.
- Assumptions and simplifications are stated (for example "no friction").
- AI explains the result; it does not invent the simulation.
- The first version supports a small set of carefully chosen simulations.

AI-27 Mistake Time Machine (solution tracing) — ADVANCED

The student types or uploads a worked solution. The system:

1. Parses the steps it can read confidently.
2. Checks each step against the relevant rule.
3. Highlights the earliest likely error.
4. Explains why that step is likely wrong.
5. Gives a similar question for the same skill.
6. Checks whether the same error appears again.

For unclear handwriting, the system asks the student to confirm its reading of each step before analysing it.

AI-28 Homework autopsy — ADVANCED

Looks at the working, not only the final answer. Possible categories: concept misunderstanding, formula choice, calculation error, sign or unit error, misreading the question, missing prerequisite. Presented as possible explanations, not psychological conclusions.

AI-29 Mistake DNA (error categories) — NEXT

Each wrong answer is tagged with a category:

Careless	the concept was known; a sign or calculation slipped
Concept	the idea is not yet clear
Speed	the idea was known; the attempt was rushed
Reading	the question was misread The student sees their category pattern over a month, with a matching fix (for example a checklist for careless errors, or a timed practice set). Teachers see class-level counts only by default.

AI-30 Full-lesson search and AI segmentation — FUTURE

Search across all uploads, and AI suggestions for recording segments.

Requires testing for accuracy before any segment reaches students.

AI-31 Chapter-to-comic studio — FUTURE

A short, curriculum-grounded comic-style explanation of a concept, in four to six panels with reusable character poses. Science and maths content must be reviewed. The first version uses reusable illustrations and text, not newly generated images for every panel.

AI-32 Branching learning adventure — FUTURE

A story where a student's choices lead to different reasoning steps. Choices are judged by predefined educational rules, not by free AI text. One short, tested adventure is the first target.

AI-33 Concept court (debate between claims) — FUTURE

Two fictional characters argue competing claims about a concept. The student decides which claim is better supported and which reasoning step is wrong. Answers are checked against approved material before use.

AI-34 Reverse exam mode (rubric feedback) — ADVANCED

The student writes an answer to a question, receives feedback against a marking guide (missing points, unclear reasoning, terminology), reads a suggested improvement, rewrites, and compares the two versions. Feedback is formative. Teacher grading stays authoritative.

AI-35 Viva simulator — ADVANCED

Conversational oral-exam practice. The AI asks a question, the student answers, the AI asks a follow-up based on the answer, and the session ends with a summary and suggested revision. Start with text. Voice follows only after text works reliably. A text fallback is always present.

AI-36 Explain-it-back mode (Feynman style) — ADVANCED

The student explains a concept in their own words, typed or spoken. The AI scores the explanation against a rubric: what was correct, what was missed, what was wrong. It then points to the exact timestamp of the teacher's recording or the note section that covers the missed point.

AI-37 Reality scanner (everyday links) — FUTURE

The student describes an everyday object or situation, or chooses one from a curated gallery. The system proposes a curriculum-linked activity, for example a bicycle and motion, or a ball and force. Camera-based recognition is a later possibility and is not assumed in the first build.

AI-38 Past-and-future reasoning challenge — FUTURE

Starts from the student's earlier explanation. Roles: the student's earlier belief, a challenge to its assumptions, and the student's revised argument supported by evidence. Final conclusions must be grounded in approved content.

AI-39 Digital twin (predicted next test) — ADVANCED

From current mastery the system builds a simulated next test and shows the student a predicted range, with the actions that could change it. Example: "Predicted 58 out of 100 at today's level. Two sessions this week may move it toward 71." Rules: the prediction is a range with uncertainty, never a guarantee; it is shown only with the evidence behind it; teachers see it only if the school allows. This feature needs a validation plan before any claim about accuracy.

AI-40 Teacher emphasis radar — ADVANCED

Finds moments in a recording where the teacher signalled importance: explicit phrases such as "this is important, remember it," a point repeated several times, or the topic with the most student doubts. The student sees a short list of key moments for the lesson. Teachers see a mirror: time spent on a topic versus interaction on it. Requires consent rules for recordings (LES-07).

AI-41 Mistake autopsy view (class) — ADVANCED

Teacher view of a shared logic break, for example "7 of 11 students swapped speed and velocity." Matched to a documented misconception library (maths and physics). One five-minute clarification is suggested.

AI-42 Teacher pattern profile ("what tends to be asked") — UNDECIDED

Summarises a teacher's past papers and emphasis signals to show question patterns, for example the share of application questions and the topics stressed. Shown to the teacher for approval before any student sees a priority list. Ethical and fairness review is required before any build.

Predictions are not guaranteed exam questions; the feature organises the teacher's own signals.

AI-43 Night recap (audio summary) — FUTURE

A short audio summary each evening: what was learned today, one fix for today's mistake, and a preview of tomorrow. Built from the student's own data and language. Audio-only mode is required (IND-02).

AI-44 Big-sibling tutor persona (name UNDECIDED) — UNDECIDED

A casual, encouraging, voice-friendly tone that sounds like an older sibling who recently took the same class. Tone is a design choice. The AI must not claim real personal experience or use fake human identity. The feature is considered only after the core tutor works.

15 Progress, learning history and motivation

PRG-01 Student progress view — CORE

Shows, in order:

1. One clear next action.
2. Concepts mastered, almost there, and needing attention (AI-13 labels).
3. Completed and pending work.
4. Recent improvements, for example "Acceleration: better this week."
5. Assessment history.

Language rule: never "weak," "average," "genius," or a rank.

PRG-02 Learning history — CORE

An authorised record of lessons opened, sessions completed, questions attempted, assessment results, repeated error patterns, teacher feedback, and learning plans. Kept for the academic year at minimum. Retention rules are in Section 25.

PRG-03 Achievement milestones — NEXT

Milestones for real progress: finishing a recovery path, showing a previously difficult concept, correcting a repeated error, finishing an assigned chapter activity. Not rewarded for time spent, click count or speed.

PRG-04 Improvement badges (growth, not rank) — NEXT

Badges show growth, for example "Acceleration: 30 to 65 percent." They do not show permanent top or bottom positions. No public leaderboards.

PRG-05 Streaks with freeze — NEXT

A daily learning streak. A streak can be paused automatically during school holidays, exam weeks and school-declared closures (IND-05). A missed day does not reset the entire streak to zero. Streaks never create pressure to study late at night; the product does not send notifications after school-set hours.

PRG-06 Daily five-minute challenge — NEXT

Each day: two items from today's topic, one revision item from earlier, one prerequisite check and one bonus. The challenge is capped at five minutes.

PRG-07 Reflection prompts — NEXT

After selected sessions, a short prompt: What did you understand? What is still confusing? Which step helped? Can you explain it in your own words?

Self-reports inform the next step but never count as proof of mastery.

PRG-08	Parent-facing study prompts	NEXT

See PAR-05.

MOTIVATION GAMES (MOT)

Games support lessons. They do not become the product. Each game uses the teacher's approved questions and changes only its wrapper. Points come from correct practice, never from screen time.

MOT-01	MCQ rush	NEXT

Timed multiple-choice questions with streaks. Helps speed and recall.

MOT-02	Match the concept	NEXT

Match terms to definitions or formulas. Helps vocabulary and formulas.

MOT-03	Rapid revision	NEXT

Five to ten mixed questions from the chapter before a test.

MOT-04	Streak challenge	NEXT

A run of correct answers. A wrong answer ends the run without penalty in points beyond the run. Encourages consistency, not speed alone.

MOT-05	Boss level (chapter unlock)	NEXT

Mixed questions across the chapter. Passing unlocks the next chapter.

Mirrors sequential mastery.

Rule after every game: a plain summary, for example "Today: 12 of 15 correct on Acceleration. Tomorrow's target: Match the concept."

16 Teacher experience

The teacher's dashboard must make daily work faster, not add administration.

The guiding principle is one teaching effort, many uses: notes, recording, session and quiz all come from the same lesson.

TCH-01 Teacher home — CORE

Shows: today's sessions and their completion; assignments awaiting review; upcoming tests; concept heat map (ANA-02); student questions needing attention; quick actions (upload, create session, assign work).

TCH-02 Class list and class detail — CORE

A list of assigned classes. Each class opens to its roster, its subjects, its published items, its review queue and its insights.

TCH-03 Roster — CORE

Students in the class with their enrolment status and activation status.

The teacher sees names here. Insights (ANA) use counts by default.

TCH-04 Publish notes and resources — CORE

Three taps: class, subject, chapter, concept, publish. A class notification is sent (COM-01). Teachers can publish from day one of the year.

TCH-05 Lesson editor — CORE

Add, delete and reorder sections. Edit text. Add examples and questions.

Attach approved resources. Preview the student view. Save drafts.

Editing a published lesson does not silently change a student's completed record: the student sees "updated," and their earlier answers stay as they were.

TCH-06 Session builder — CORE

See SES-03.

TCH-07 Homework and quiz creator — CORE

See ASM-02. Uses the question bank (ASM-08) or AI drafts (AI-22).

TCH-08 Auto-marking and review queue — CORE

See ASM-04 and ASM-05. The queue prioritises borderline cases so the teacher reviews a small number of answers in depth.

TCH-09 Class announcements — CORE

Post an update to the class (for example "Physics test moved to Thursday").

Announcements reach only the intended class. Parents receive them only when the school allows (COM-02).

TCH-10 Teacher settings — CORE

Profile, notification preferences, default hint rules, Socratic default (AI-11), and whether the teacher's class uses recordings (SCH-12).

TCH-11 Class teacher role — CORE

Extra rights for one class: manage the class space (STU-11), post class updates, and send the parent progress report for the class (PAR-03).

Subject teachers see their own subject only.

TCH-12 Reteaching decision — CORE

The teacher uses insights to choose: reteach now, assign practice, or move on. Each choice can be recorded as an intervention (ANA-04) so its effect can be checked later.

17 Teacher insights and analytics

ANA-01 Class completion view — CORE

Shows, for the chosen session or assignment: how many students started, finished, and submitted. Names are shown only in the roster. This view shows counts.

ANA-02 Concept heat map — CORE

A grid of concepts against the class. Each cell shows the share of the class answering correctly, coloured green (most correct), peach (mixed) or red (most stuck). A stuck concept is visible before the next test.

Example: Acceleration shows red, 60 percent of answers wrong.

Note: the colours use the palette in Section 4; red is used only here, for warnings, and is paired with a text label.

ANA-03 Doubt clustering — ADVANCED

Student

AI questions are tagged to concepts. The teacher sees counts by concept: "17 doubts on Acceleration this week, 3 on Velocity, none on Distance." Default: counts only. Names are hidden unless the school and teacher choose named mode [OPEN] (Decision 6). Students should know how doubts are used. Reason: if students believe the teacher can see who asked, they stop asking.

ANA-04 Intervention record and follow-up — ADVANCED

Each intervention (reteach, practice, move on) is logged with its date, its target concept and the follow-up result. Over time this shows which interventions helped. Causation is not claimed from a single case.

ANA-05 Mistake autopsy class view — ADVANCED

See AI-41.

ANA-06 Class trend view — NEXT

Completion, assessment distribution and change between assessments, by concept. Shows trends, never public rankings.

ANA-07 Students who may need extra support — NEXT

A short list, based on several recent signals (not one test). Shown with reasons, and the teacher decides what to do. Names shown only to the assigned teacher.

ANA-08 **Teacher mirror (teaching time versus interaction)** **ADVANCED**

Compares teaching time on each topic with student interaction on it. Used for self-reflection by the teacher, never for inspection by the school.

18 Parent experience

Parents should understand their child's learning without becoming the child's teacher, and without constant surveillance.

PAR-01 Parent home — CORE

Answers five questions: What is my child learning now? What is complete?

Where might they need help? What is the teacher doing? What can I do without adding pressure?

Shows: child's name, class and school; current learning focus; assigned work with deadlines; completed activities; the latest assessment result; concepts needing practice; teacher updates; one suggested support action.

PAR-02 Child view — CORE

Subjects and chapters, sessions completed or pending, and quiz results in plain form. Data is only for the verified child.

PAR-03 Progress summary (weekly or on event) — CORE

Written in plain language, for example: "Your child can recall the definitions of speed and velocity. They need more practice applying them to word problems. A short practice session has been assigned."

The summary labels its parts: observed result, AI interpretation, teacher confirmed, suggested action. It never presents an inference as a medical, psychological or permanent-ability diagnosis.

Frequency is [OPEN] (Decision 10).

PAR-04 Assignments and deadlines — CORE

Assignments with status and due dates. Read-only for parents.

PAR-05 Support suggestions ("Ask this today") — NEXT

One simple suggestion a parent can use without teaching the subject, for example: "Ask Aarav: what is the difference between acceleration and velocity? The person who explains it remembers it best."

Rules: suggestions respect the parent's time and the child's independence, and never replace the teacher's plan.

PAR-06 Linked children and account settings — CORE

Lists all verified children. A parent can switch between them. Shows the verification status of each link and lets the parent request a correction through the school.

PAR-07 Plain-text message fallback (WhatsApp or SMS) — NEXT

Key updates can be sent by WhatsApp or SMS for parents who do not open the app. Example: "Aarav: Science homework pending (due tomorrow). Quiz 4 of 5. Today's revision: Acceleration, 20 minutes."

Rule: messages are sent only to the verified parent number and contain the minimum needed. Provider and cost [OPEN].

PAR-08 Privacy boundaries (hard rules) CORE

Parents cannot see:

- Private conversations with Saathy (by design, not by setting).
- Other students' data.
- Public rankings.
- Anything about children they are not linked to.

Parents can see only the verified child's approved progress and tasks.

PAR-09 Parent meeting preparation ADVANCED

Before a parent-teacher meeting, the parent can review the child's data and practise questions to ask, for example: "Where is my child getting stuck in Acceleration?" The child can also prepare a short, evidence-based account of their progress. Uses only the child's approved data. Teachers are not shown the parent's practice sessions.

Note: this is a rehearsal aid. It does not simulate or script the teacher.

19 School administration

SCH-03 Academic year — CORE

Create the year, copy the previous year's structure, add new students and dates. Exam weeks, holidays and school closures are entered here and feed the calendar awareness layer (IND-05).

SCH-04 Classes and sections — CORE

Create classes and sections such as 9-A and 9-B, and any streams. Two-minute setup through the wizard.

SCH-05 Subjects — CORE

Add subjects per class. Link to the board's chapter structure (LES-10).

SCH-06 Teacher assignments — CORE

Map teacher x subject x class. Mid-year changes are common in India, so reassignment must update access immediately and keep history.

SCH-07 Student enrolment — CORE

CSV upload with validation: duplicate roll numbers, missing parent numbers, wrong class names. Preview before confirming. Generates activation codes and a distribution sheet for the school to hand out.

Open: whether the admin, the class teacher or students themselves upload the list (Decision 3).

SCH-08 Parent linking management — CORE

Which parent is linked to which child; phone verification status; cases of one parent with several children; guardian cases. Admin can correct a link with a logged reason.

SCH-09 Invitations — CORE

Track which teachers, students and parents have accepted. Resend or cancel an invitation. Expired invitations are marked clearly.

SCH-10 Admin home — CORE

Structure and health: classes, teachers, enrolment counts, activation status, parent-linking status, published resources. No student rankings and no individual AI chats.

SCH-11 Content library permissions — CORE

Decide which resources can be shared across classes and which are restricted.

SCH-12 School settings — CORE

Recording use (on or off); doubt visibility (counts or named); parent update frequency; whether AI is enabled for students; whether competition features are enabled (default: no leaderboards); assessment rules.

SCH-13 Academic reports — NEXT

Completion trends, subject health and resource coverage across the school.

Used to support teaching and planning, never to rank or punish.

SCH-14 Admin activity log — CORE

Records sensitive administrative actions: enrolment changes, link changes, permission changes. Viewable by the platform administrator.

20 Notifications and communication

COM-01 Student and class notifications — CORE

Triggers: new notes or session published; session released; assignment due soon; teacher announcement. Each notification opens the exact item.

COM-02 Teacher announcements — CORE

Covered in TCH-09. Recipients: the class only.

COM-03 Parent notifications — CORE

Triggers: new assignment; upcoming deadline; a published progress summary; an important teacher announcement; a teacher-requested follow-up. Parents can choose frequency within school rules.

Avoid alerts for every click, question or minor score change.

COM-04 Reminders — NEXT

One reminder per item, at a school-set time, never after school hours.

COM-05 Quiet hours — CORE

No notifications between the school-set quiet hours. Default quiet hours [OPEN].

COM-06 Delivery failure — CORE

If a notification cannot be delivered, the item still appears in the app's notification centre. Failures are logged without student personal data.

COM-07 Notification centre — CORE

A list of recent notifications for the user. Read and unread state.

COM-08 Direct messaging — NOT IN FIRST RELEASE

Full parent-teacher or student-student messaging is not assumed. Teacher replies to doubts (AI-12) and class announcements (TCH-09) cover the first need. A messaging system is [UNDECIDED].

21 India-specific layer

The following features address conditions the sources describe for Tier-2 and Tier-3 cities and schools. The sources describe these as hypotheses that still need field research (see Section 30). Tier-1 students also use shared devices and unreliable connections; the design should not assume that Tier-2 and Tier-3 students lack all technology.

IND-01 Offline-first study and sync

NEXT, access requirement

The student taps "Download today's session" when connected. Download is text-first, with compressed images and optional low-resolution video, about 2 to 3 MB for a typical session. At home without a connection, the cached session opens. Quiz answers are saved on the device and sent when the connection returns. The teacher dashboard updates after sync.

Rules:

- Downloading is the student's explicit action; the app does not download large files without asking.
- Conflicts are resolved by timestamp: the latest submitted answer for a given question is kept, and the earlier one is logged.
- A pending submission shows a visible "waiting to send" status.

Note: S4 calls offline access non-negotiable for India; S3 calls it a later idea. Decision 12 covers the first-release scope.

IND-02 Data saver and audio-only mode

NEXT

Each resource shows its size ("Session 4: 2 MB"). Recordings offer audio-only playback. Lightweight images and text by default.

IND-03 Language layers

CORE for UI, NEXT for AI

Three separate things:

1. Interface language: the app's labels and menus (English first; a regional language later [OPEN]).
2. AI answer language: the language the student asks in is the language of the reply.
3. Bridge explanations: English notes stay as they are, and the student can ask for a Hinglish or regional explanation with the same technical terms in English.

Rule: textbook and exam terminology stays accurate. An informal explanation is followed by the standard definition expected in the exam.

IND-04 Low-end device support

CORE

Lightweight web app or progressive web app. Target: a usable screen within about two seconds on a mid-range phone with a slow connection. Limited animation.

Compressed images. Works on phones with little memory.

Measure before claiming a number [OPEN: test devices].

IND-05	Calendar awareness	NEXT

The system knows school holidays, exam weeks and festival dates (from the academic year, SCH-03). Effects: deadlines shift away from holidays; streaks pause (PRG-05); notification timing adjusts. This is a small feature with a large effect on whether the product fits the school calendar.

IND-06	Parent and student shared-device privacy	CORE

See AUTH-09. Sign out and profile chooser prevent one family member from opening another's account.

22 Complete cross-role user journeys

Each journey lists the steps, the screen, what the user does, what the system does, and the next state. Names and data are fictional.

JOURNEY A School setup to first class (Admin)

1. Admin opens the landing page and chooses school registration (AUTH-01).
2. Admin completes the wizard (SCH-01): profile, academic year, classes 9-A and 9-B, subjects, teacher assignments.
3. Admin uploads the student list (SCH-07). The system validates the file and lists two duplicate roll numbers. Admin fixes them and confirms.
4. System generates activation codes and a distribution sheet.
5. Admin invites teachers (SCH-09). Teachers accept.

State after: the class workspace for 9-A is live.

JOURNEY B Student activation and first day (Student and parent)

1. Student enters school, roll number, class and section (AUTH-03).
2. Student enters the activation code. System rejects a wrong code and shows "Check the code or ask your school."
3. Parent receives a code by phone and verifies it (AUTH-05).
4. Student sets a 4-digit PIN (AUTH-04).
5. Welcome screen (ONB-01). Student sees Motion, Session 3 (scheduled).

State after: student and parent are linked and active.

JOURNEY C Teacher publishes a lesson and session (Teacher)

1. Teacher uploads the Motion notes (LES-02) and tags them to Distance, Speed and Velocity, and Acceleration (LES-03).
2. Teacher uploads the lesson recording and marks an Acceleration segment (LES-07).
3. Teacher builds Session 3 from the template (SES-03), adds slides and five MCQs, and sets the release for 18:00 today.
4. Teacher previews the student view and publishes (TCH-04).
5. Students receive a notification (COM-01).

State after: Session 3 is scheduled; it appears on student home at 18:00.

JOURNEY D Student completes a session and gets a next step (Student A)

1. Student A sees "Session 3, 15 minutes, start?" on home (STU-01).
2. Completes slides and questions in the player (SES-02).
3. Scores 5 of 5 on the quiz. Summary shows completion and next step.
4. System updates the concept map: Speed and Velocity "Almost there" moves toward mastered; the estimate needs more varied evidence (AI-08).
5. System offers a harder Speed and Velocity set (AI-09).

State after: Student A has the next step; the teacher sees completion counts.

JOURNEY E Student struggles and receives a recovery path (Student B)

1. Student B scores 2 of 5. "Why this was wrong" shows the Acceleration explanation and the relevant slide (AI-04).
2. System checks the prerequisite chain (AI-07): velocity questions were answered correctly, so the gap is in Acceleration.
3. Student B taps "Explain another way" and chooses the everyday-analogy mode (AI-05).
4. Student B works three worked examples and three practice questions (AI-06, AI-09). Hints are available (AI-10).
5. Student B asks Saathy a question and receives a grounded answer with source (AI-01).
6. Mini check passes. Estimate updates (AI-08). Next step: application questions.

Also: the answer error category is recorded (AI-29) as "Concept."

JOURNEY F Student who missed the class catches up (Student C)

1. Home shows the card: "You missed Motion yesterday. Catch up in 20 minutes?"
2. Student opens the 7-minute Acceleration recording segment (LES-07).
3. Reads the notes for the concept (LES-03).
4. Completes a mini session and a five-question check (SES-01).
5. Passes. The card disappears. Class pace is restored.

Failure path: if the check fails, the system recommends the prerequisite check (AI-07) and a teacher question (AI-12).

JOURNEY G Teacher reviews class patterns and intervenes (Teacher)

1. Teacher home shows the concept heat map: Acceleration in red (ANA-02).
2. Teacher sees doubt counts by concept, with names hidden (ANA-03).
3. Teacher opens the mistake autopsy: 7 of 11 students swapped speed and velocity (AI-41).
4. Teacher decides to reteach for 10 minutes and logs the intervention (TCH-12, ANA-04).
5. Next day's check shows Acceleration moved from red to peach.

State after: intervention recorded with its follow-up result.

JOURNEY H Parent sees progress and a suggestion (Parent)

1. Parent signs in and sees the child's home (PAR-01).
2. Reads the weekly summary (PAR-03): Session 3 complete, quiz 4 of 5, one area to practise.
3. Reads the suggestion "Ask this today" (PAR-05).
4. Parent cannot open the child's AI conversation. The screen says so in plain words (PAR-08).
5. Parent receives a plain-text WhatsApp message if the school enables it (PAR-07).

State after: parent is informed; no private chat was exposed.

JOURNEY I Teacher changes a class assignment mid-year (Admin)

1. Admin opens teacher assignments (SCH-06) and moves a teacher from 9-B to 9-A for Physics.
2. System updates the teacher's access immediately. Old access is removed.
3. History keeps the earlier assignment for reporting.

Failure path: if the teacher is editing a draft for 9-B, the draft is kept and the teacher is told that they no longer have publishing rights for 9-B.

23 Feature dependency map

A feature cannot work well unless the features it depends on exist.

Feature	Depends on
Class workspace (LES-01)	School setup (SCH-01), class and subject structure (SCH-04, SCH-05)
Notes by concept (LES-03)	Resource upload (LES-02), concept structure (LES-10 or teacher-created concepts)
Guided session (SES-02)	Session builder (SES-03), notes and questions, timed release (SES-05)
Quiz results	Session player (SES-02) or assignment (ASM-02)
Concept map (AI-13)	Evidence (AI-08), concept graph (AI-03)
Study plan (AI-06)	Evidence, concept graph, notes, recordings
Prerequisite gap (AI-07)	Concept graph (AI-03), per-concept evidence
Ask Saathy (AI-01)	Concept tags (LES-03), notes, class context
Heat map (ANA-02)	Quiz and assignment results across the class
Doubt clustering (ANA-03)	Ask Saathy questions tagged to concepts

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Parent   summary (PAR-03)      Approved evidence; teacher approval for
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any teacher-written comment

Notifications (COM-01)	Publishing (LES-04), release (SES-05)
Offline sync (IND-01)	Session download, local storage, sync queue
Learning Remix (AI-21)	Question bank (ASM-08), teacher sample approval
Digital twin (AI-39)	Evidence history, question bank, validation

Key chain (the core):

Notes (LES-03) > Session (SES-02) > Quiz (ASM-07) > Evidence (AI-08) > Concept map and plan (AI-13, AI-06) > Teacher heat map (ANA-02) > Intervention (TCH-12) > Parent summary (PAR-03).

Cloud mascot dependency: the mascot (Section 4) reacts to events from the player and the AI. It does not decide what a student should study. Educational logic lives in Sections 11 to 13.

24 Conceptual data model

This section describes the information the product must keep, in plain terms.

Exact tables, columns and identifiers are a technical decision (Section 30).

Entities:

School	a school; has academic years, classes, staff
AcademicYear	start and end dates, holidays, exam windows
Class	a class (for example 9) within a year
Section	a section within a class (for example A)
Subject	a subject taught to a class
TeacherAssignment	teacher x subject x class, with start and end dates
Membership	a user's role in a school
User	a person with an account (student, teacher, parent, admin)
StudentProfile	student details tied to a section and roll number
ParentLink	parent to child, with verification status
Chapter	a chapter of a subject
Concept	an idea within a chapter ConceptPrerequisite link from a concept to the concept it depends on
Resource	a note, PDF, slide set, worksheet or link
ResourceConceptTag	which concepts a resource supports
Recording	a lesson recording with concept segments
Session	a guided session; has release and deadline times
SessionSection	one slide or step inside a session
Assignment	homework or test
Question	a question with difficulty and format
QuestionOption	answer options for an objective question
Attempt	a student's attempt at an assignment or session
Response	an answer inside an attempt
Feedback	teacher or approved AI feedback on a response
Evidence	per-concept record: attempts, hints, time, outcome
MasteryEstimate	working estimate per student per concept, with uncertainty
LearningPlan	a plan with steps and status
Intervention	teacher action with a target concept and follow-up
Announcement	class-level message
Notification	message to a user, with delivery status
ProgressSummary	parent-facing summary, with its approval record
DoubtRecord	a student question tagged to a concept, with privacy level
AuditEvent	record of a sensitive administrative action

Relationships in plain words:

- A school has many academic years, classes and staff memberships.
- A class has many sections and subjects. A teacher assignment connects a teacher, a subject and a class.

- A subject has chapters; a chapter has concepts; concepts link to each other through prerequisites.
- Resources, sessions and questions attach to concepts.
- A student's attempts create responses, which create evidence per concept.
- Evidence updates the mastery estimate for that student and concept.
- A parent link connects one parent user to one or more student profiles.
- Parent summaries are built only from approved evidence and are stored with the approval record.

Data rules:

- Every student record belongs to exactly one school.
- Every query is filtered by school, and then by role and relationship.
- Changing a published lesson does not rewrite past responses (TCH-05).
- Deleting content is not allowed once it has student responses; archive it.

Checklist:

[ ] Entity list reviewed by the developer.

[ ] Relationships checked against Section 23.

[ ] Retention rules agreed (Decision 11).

25 Security, privacy and responsible AI

25.1 Security basics (SEC-01 to SEC-04) [CORE]

SEC-01 Server-side permission checks on every request. Client-side role labels never grant access.

SEC-02 Secure password storage, time-limited reset links, rate limits on sign-in and activation attempts (AUTH-03, AUTH-04).

SEC-03 Secure transport for all traffic. Credentials are never placed in browser-visible code. API keys stay on the server.

SEC-04 Error messages reveal no account details. Logs contain no personal student data.

25.2 Data minimisation and retention (SEC-05) [CORE]

Collect only what each feature needs. Define retention periods for learning history, AI interaction logs and notifications [OPEN: Decision 11]. Support account correction and deletion requests through the school.

25.3 Parent-student access (SEC-06) [CORE]

A parent sees only verified, linked children. Link changes are logged (SCH-08, SCH-14).

25.4 Child-safety and age-appropriate interaction (SEC-07) [CORE]

- Ask Saathy answers within school subjects and uses age-appropriate language.
- Off-topic or unsafe questions are redirected politely and, where the content concerns a student's wellbeing, passed to a teacher or a trusted adult, not answered as a learning question.
- The product does not offer advice on self-harm, substances or other sensitive personal topics; it directs the student to a trusted adult.

Note: the exact escalation process and staffing for this are a school and policy decision [OPEN].

25.5 Responsible AI (SEC-08) [CORE]

- Ground school answers in approved material where possible (AI-01, AI-02).
- Show uncertainty. Never invent citations, page numbers or teacher remarks.
- Keep AI drafts separate from published content. Teacher approval is required for any content that reaches students.
- Avoid medical or psychological diagnosis and permanent labels.
- Treat instructions inside uploaded files as untrusted text, not as commands to the AI (prompt injection defence).
- Provide a fallback when the AI service fails (Section 27).
- Deterministic calculations for numerical simulations (AI-26).

25.6 Evidence integrity (SEC-09) [CORE]

Keep these separate in the database and in every display:

Activity completed	the student opened or finished something
Answer correct	a single question was right
Concept demonstrated	evidence across formats and time supports it

Teacher-verified	a teacher has confirmed the concept AI-inferred difficulty a system estimate, shown with uncertainty A weak inference is never shown as a fact.

25.7 Features that need a privacy review before build (SEC-10)

The following features use behavioural or sensitive patterns. Each needs a written privacy and fairness review, school consent rules, and a visibility setting that defaults to the most private option:

- Avoidance radar (STU-A02 below): detects topics a student repeatedly skips.
- Teacher pattern profile (AI-42).
- Named doubt visibility (ANA-03).
- Frustration-aware responses (AI-17).
- Digital twin predictions (AI-39).
- Teacher emphasis radar with recording analysis (AI-40).

Avoidance radar detail (STU-A02): the system watches for repeated skipping, very short time on a concept, or quick hint requests on one concept across several weeks. The student sees a gentle note and a small starting task. It is never shown to parents, and teachers see it only if the school opts in.

Status: UNDECIDED.

25.8 Camera and microphone (SEC-11)

Emotion detection from camera or microphone is excluded from scope.

Voice input for questions (related to AI-35) is a separate feature and is opt-in per session.

25.9 Applicable law and school policy

Compliance with Indian data-protection requirements and school child-data policies must be confirmed by the team with qualified advice. No compliance status is claimed in this document.

26 Screen inventory and interface states

Each screen below has: purpose, user, primary action, and states (empty, loading, error, success). Shared states follow Section 4.7.

Public and access screens

```
SCR-01 Landing page            visitors       primary: Sign in or Register school
SCR-02 School registration     school rep.    primary: Start setup
SCR-03 Sign-in                 all users      primary: Sign in
SCR-04 Activation              students       primary: Activate
SCR-05 PIN set and sign-in     students       primary: Continue
SCR-06 Account recovery        all users      primary: Send reset link
SCR-07 Access denied           any            primary: Go home
```

SCR-08 Profile chooser shared devices primary: Pick profile

Student screens

```
SCR-10 Home                    student        primary: Next action
SCR-11 Today's plan            student        primary: Start item
SCR-12 Subject list            student        primary: Open chapter
SCR-13 Chapter and concepts    student        primary: Open concept
SCR-14 Concept detail          student        primary: Study or Ask
SCR-15 Session player          student        primary: Next slide
SCR-16 Session summary         student        primary: Next step
SCR-17 Resource viewer         student        primary: Back to concept
SCR-18 Recording player        student        primary: Play segment
SCR-19 Assignment list         student        primary: Open assignment
SCR-20 Assignment answer       student        primary: Submit
SCR-21 Ask Saathy              student        primary: Ask
SCR-22 Study plan              student        primary: Start step
SCR-23 Concept map             student        primary: Open concept
SCR-24 Progress                student        primary: Next action
SCR-25 Profile and settings    student        primary: Save
SCR-26 Notifications           all users      primary: Open item
```

Teacher screens

```
SCR-30 Teacher home            teacher        primary: Create or review
SCR-31 Class list and detail   teacher        primary: Open class
SCR-32 Roster                  teacher        primary: View student
SCR-33 Resource library        teacher        primary: Upload
SCR-34 Upload and tag          teacher        primary: Publish
SCR-35 Lesson editor           teacher        primary: Save or publish
SCR-36 Session builder         teacher        primary: Schedule
SCR-37 Assignment builder      teacher        primary: Assign
SCR-38 Review queue            teacher        primary: Open item
SCR-39 Heat map                teacher        primary: Open concept
SCR-40 Doubt counts            teacher        primary: Open concept
SCR-41 Interventions           teacher        primary: Log intervention
SCR-42 Announcements           teacher        primary: Post
SCR-43 Teacher settings        teacher        primary: Save
```

Parent screens

```
SCR-50 Parent home            parent          primary: Read summary
SCR-51 Child view             parent          primary: Open subject
SCR-52 Progress summary       parent          primary: Read or share
SCR-53 Assignments            parent          primary: Open item
SCR-54 Support suggestion     parent          primary: Done
SCR-55 Linked children        parent          primary: Switch child
SCR-56 Parent settings        parent          primary: Save
```

School admin screens

```
SCR-60 Admin home             admin           primary: Setup or review
SCR-61 School setup wizard    admin           primary: Next step
SCR-62 Academic year          admin           primary: Save
SCR-63 Classes and sections   admin           primary: Add class
SCR-64 Subjects               admin           primary: Add subject
SCR-65 Teacher assignments    admin           primary: Assign
SCR-66 Student enrolment      admin           primary: Upload list
SCR-67 Parent links           admin           primary: Review link
SCR-68 Invitations            admin           primary: Resend
SCR-69 School settings        admin           primary: Save
```

SCR-70 Admin activity log admin/platform primary: Filter

Checklist for each screen (apply before build):

[ ] Empty state written and reviewed.

[ ] Loading state uses skeleton, not a blank page.

[ ] Error state explains what happened and what to do next.

[ ] Success state has a next step.

[ ] Permissions tested for each role.

[ ] Works on a small phone screen.

27 Error handling and edge cases

Principle: work is never silently lost. Each failure gives a plain message and a safe next step.

Case	Expected behaviour
No session assigned today	Empty state: "Nothing assigned today. Explore your subjects."

```
Teacher   has not uploaded material   Chapter shows "Notes coming soon" and
```

the teacher's name; no fake content.

Session scheduled, not yet open	Shows "Opens at 18:00 today."
Connection lost during a quiz	Answers saved on the device; "Waiting to send" status; submits on reconnect.
Assignment submission fails	Kept on device; retry button; teacher is not told a submission exists until it does.
File cannot be processed	Kept as upload; "Preview not available"; can still be published.
AI returns invalid or unsupported	Discarded; fallback to saved notes; the
answer	student sees "I could not check that. Here is your teacher's note."
AI service unavailable	Saved notes for the concept, retry button, and no error code shown to the student.
Ambiguous wrong answer	System says the reason is unclear and offers two possible explanations, not one.
Repeated struggle on one concept	Suggests teacher help after two explanation attempts (AI-05).

```
Parent   invitation expired           Marked expired; school admin can resend.
Parent   linked to wrong student      Parent reports it; admin corrects the link
```

with a logged reason (SCH-08).

```
Teacher   opens unauthorised class    Access denied page (SCR-07). No data shown.
```

User opens protected URL directly	Server rejects; access-denied page.
School changes teacher assignment	Access changes immediately; drafts kept and the teacher is told what changed.
Lesson edited after students start	Student sees "updated"; earlier answers kept as they were (TCH-05).
Notification cannot be delivered	Item stays in the notification centre; failure logged without personal data.

```
Student   signs out on shared device   Profile chooser appears; no data remains
```

visible.

Activation code wrong or reused	"Check the code or ask your school." Five failures lock for 15 minutes.

```
Student   forgets PIN                Parent or teacher approves a reset.
```

Recording upload interrupted Upload resumes from where it stopped if the platform supports it [OPEN: technical].

Checklist:

[ ] Each case above has a test (Section 29).

[ ] No failure path shows a raw technical error to a student.

28 Implementation phases and status definitions

28.1 Status definitions

CORE	required for the central loop. Without it the product is not the product.
NEXT	valuable after the loop works; can follow in the same term.
ADVANCED	needs extra engineering, content, validation or review.
FUTURE	a longer-term possibility, kept for the record. UNDECIDED needs a team decision before any build.
EXISTING	would mean the feature is demonstrably built. No feature is marked EXISTING in this document, because no implementation evidence was supplied.
PROPOSED	described in the sources but not confirmed.

28.2 Phase 1 (six-hour prototype scope, suggested)

Objective: one complete, convincing loop, not many disconnected screens.

- [ ] Landing and sign-in screen with the logo, palette and mascot (SCR-01, SCR-03).
- [ ] Demo school with Class 9-A, one subject (Physics) and one chapter (Motion).
- [ ] Three role home screens: student, teacher, parent (SCR-10, SCR-30, SCR-50).
- [ ] Teacher publishes one note set and one session (LES-02, LES-03, SES-03).
- [ ] Student plays the session with a five-question quiz (SES-02, ASM-07).
- [ ] Two seeded student profiles with different results (Student A and B).
- [ ] One grounded AI answer with source, plus a fallback (AI-01, AI-15).
- [ ] One prerequisite recommendation from the gap logic (AI-07).
- [ ] Teacher heat map for the class (ANA-02).
- [ ] Parent summary in plain language (PAR-03).
- [ ] Mascot states used in the right places (Section 4.4).
- [ ] End-to-end demo run and recorded.

Demo data must be labelled as simulated. The demo must not claim real learning results.

28.3 Phase 2 (functional pilot)

Real accounts and activation (AUTH-01 to AUTH-07); school setup (SCH-01 to SCH-09); resource and session persistence; assignments and auto-marking; parent linking; teacher review queue; basic notifications; teacher-approved AI drafts (AI-22); question bank (ASM-08); basic offline download (IND-01) if the team decides so.

28.4 Phase 3 (intelligent layer)

Concept graph (AI-03); evidence and mastery estimate (AI-08); adaptive practice (AI-09); study plan (AI-06); recovery path; concept heat map with intervention records (ANA-04); evidence-based parent summaries (PAR-03).

28.5 Phase 4 (signature experiences)

Remix (AI-21), mistake tracing (AI-27), explain-it-back (AI-36), viva (AI-35), reverse exam (AI-34), reality scanner (AI-37), comic studio (AI-31), adventure (AI-32), concept court (AI-33), digital twin (AI-39).

Prioritise by educational value, cost and reliability, not by novelty.

28.6 Phase 5 (scale and adoption)

Bulk onboarding; more grades and boards; more languages; low-bandwidth improvements; wider offline; school-to-school record transfer with consent; deeper analytics; integrations with school systems where practical.

28.7 Build discipline

- One loop first. Do not start Phase 4 features before the Phase 1 loop works.
- Each feature ships with its acceptance criteria (Section 29).
- Any claim about learning results needs real user testing first.

29 Acceptance criteria checklist

Accounts and permissions

[ ] A student cannot open a teacher or admin page by changing the URL.
[ ] A parent sees only children linked and verified to them.
[ ] A teacher sees only assigned classes and student work.
[ ] Signing out prevents access to protected pages.
[ ] Failed sign-in or activation does not reveal account details.
[ ] A roll number alone never signs a student in.

Learning workflow

[ ] A teacher can publish notes to the intended class only.
[ ] An authorised student can open a published session at its release time.
[ ] The student can answer and submit a supported quiz.
[ ] The system records the answer and its result.
[ ] The student sees an appropriate next step.
[ ] Completing a session requires a quiz or check attempt.
[ ] The teacher can view class-level results for an assigned class.
[ ] The parent sees only the permitted summary.
[ ] The app separates activity completed from concept demonstrated.

AI behaviour

[ ] Answers use approved school material where available.
[ ] Unsupported questions do not produce fabricated sources.
[ ] AI drafts stay unpublished until a teacher approves them.
[ ] Numerical simulations use deterministic calculations.
[ ] Unclear handwriting can be confirmed by the student before analysis.
[ ] AI failure does not lose saved work.
[ ] No mastery label is shown from one correct answer.

Offline and devices (when IND-01 is in scope)

[ ] A downloaded session opens without a connection.
[ ] Answers taken offline are sent once when the connection returns.
[ ] Conflicting answers keep the latest and log the earlier one.

Experience

[ ] The primary student action is easy to find on each screen.
[ ] The teacher can publish a simple session in a few steps.
[ ] The parent understands the progress summary without help.
[ ] Text is readable on a small phone; actions give clear feedback.
[ ] The mascot never blocks a lesson or flashes constantly.
[ ] Errors say what happened and what to do next.
[ ] Reduced-motion setting removes mascot animation.

Validation before any claim

[ ] Students complete a guided session on their own.

- [ ] A personalised explanation improves performance on a fresh question.
- [ ] Teachers spend less time than in their current workflow.
- [ ] Teachers trust AI drafts after review.
- [ ] Parents understand progress summaries.
- [ ] The product works on the devices and connections students actually use.
- [ ] The learning loop does better than sharing a PDF or video alone.

30 Open decisions and risks

30.1 Decisions to make (do not guess)

#	Decision	Options	Status
1	Scope of hackathon prototype	Single loop vs wider	OPEN
2	Student login and verification	Activation code, parent OTP,	OPEN

admin approval

3	Who uploads the student list	Admin, class teacher,	OPEN

self-join with code

4	Minimum evidence before a gap is	Question count and types	OPEN

suggested

5	Mastery tracking	Overall %, per concept,	Per

dynamic (per concept concept probable **likely)**

6	Doubt visibility	Counts only or named	OPEN
7	Any parent access to AI chats	None (current direction) or a	OPEN

written school exception

8	Session completion rule	Slides viewed, quiz, or both	PROPOSED

(quiz)

9	First pilot school	Own school, partner, other	OPEN
10	Parent update frequency	Daily, weekly, on event	OPEN
11	Data retention and deletion	Periods and rules	OPEN
12	Offline access in first release	Yes or later	OPEN
13	Recording consent rules	School policy, student and	OPEN

guardian consent

14	Technology stack	React or Next.js with	OPEN

Supabase/PostgreSQL, or other

15	Product name	VidhyaSaathy or ClassOS	OPEN
16	Tagline	Learn . Grow . Together or	OPEN

another

17	Signature AI feature first	See Section 14	OPEN
18	Persona name for the tutor	See AI-44	OPEN
19	Interface font	To choose	OPEN
20	Quiet-hours default	To set	OPEN

Note on numbering: this table is the numbering used throughout this document.

The Team Guide (S3) uses different numbers; where the two differ, this table is the working version.

30.2 Risks and mitigations

Risk	Why it matters	Mitigation
Teachers do not use it	No content, no loop	Publish in a few taps; reuse last year's notes; measure teacher time in pilot
Students skip the work	No evidence	Quiz rule; short sessions; visible next step; teacher visibility
Students open all slides at once	Looks like progress, no learning	Sequential unlock; question per slide
Parents do not engage	Weak home support	Plain summaries; WhatsApp fallback; one suggestion
AI gives a wrong answer	Harm to learning and trust	Grounding; source display; teacher approval; honesty layer
Privacy breach	Serious harm to children	Server-side permissions; minimal data; audit log; review
School procurement is slow	Pilot delayed	Pilot with one school first; clear value summary
Poor internet and shared devices	Students cannot use it	Light pages; offline (IND-01); profile chooser
Teacher workload	Adoption fails	Reuse; AI drafts for review only
Gamification becomes a distraction	Time spent, not learning	Games use approved questions; no leaderboards
Students game the system	Fake progress	Evidence across

formats; checks required

```
AI dependence              Students stop thinking          Socratic mode;
```

hints ladder; explain-it-back

```
Behavioural features feel  Trust loss, avoidance of the    Privacy review
like surveillance          app                             (SEC-10); opt-in;
```

counts not names

```
Claims outrun evidence     Credibility loss                Validation
```

before claims (Section 29)

30.3 Evidence gaps to close before pitching outcomes

- No survey data on how often students face fragmented resources. The founder's observations are hypotheses.
- No field data on Tier-2 and Tier-3 device access, connectivity, teacher digital skill or school purchasing.
- No validation of the four performance bands or the mastery estimate.
- No review of recording consent or child-data rules for the pilot school.

The team should collect these with a small study before claiming any learning effect.

31 Glossary

Activation code One-time code a student uses to join their class account.

Adaptive practice Practice that changes difficulty based on recent answers.

Ask Saathy The AI question feature for students. Saathy is the AI helper's name in the brand.

Class workspace The online space for one class, mirroring the offline class.

Concept One idea inside a chapter (for example, Acceleration).

Concept map A visual tree of a chapter's concepts with the student's status for each.

Concept heat map The teacher's grid of concepts against the class.

Evidence A recorded learning action: an answer, a session completion, a hint request.

Guided session A school-prepared, fixed-structure learning experience, completed at home.

Grounding Tying an AI answer to approved school material.

Hint ladder The ordered hints from a pointer to a full explanation.

Mastery estimate A working, uncertain estimate of understanding from evidence.

Prerequisite A concept that must be understood before another.

Question bank The school's library of approved questions.

Release time When a session becomes visible to students.

Socratic mode The AI guides with questions instead of giving the answer.

Teacher approval The required step before AI content reaches students.

32 Final product definition

VidhyaSaathy is a school-connected learning workspace for each class. It keeps the school's support working after the school bell: teachers publish approved notes, recordings, guided sessions and homework; students follow a clear next step and ask questions in the context of their own school's material; teachers see class-level patterns and decide what to reteach; parents see approved, plain-language progress and one useful way to help.

Its foundation is not the number of AI features. It is the connection between school-approved content, student evidence, a grounded next step and a teacher's decision. The cloud mascot gives the product a friendly face, the guided session gives it structure, the AI gives it personalisation, the teacher heat map gives it insight, and the parent summary gives it shared understanding.

Product principle: One classroom. Different learning paths. One connected support system.

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